

BMCS2133

IMAGE PROCESSING

Welcome onboard



Course Learning Outcome



CLO1

Analyse the theories and different techniques in digital image processing.



CLO2

Practice suitable image processing techniques and algorithms to solve a given problem.



CLO3

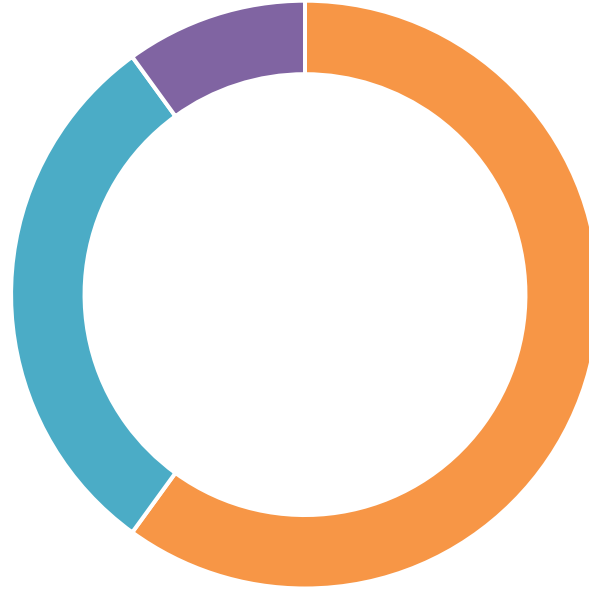
Present solution by utilizing the image processing techniques and tools to address real-world problems.



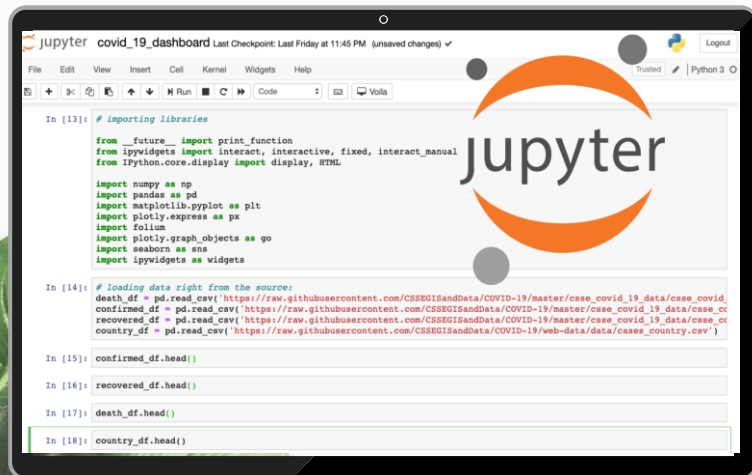
Introduction

This course provides students with the knowledge of the **digital image processing techniques** and how digitized images are represented and manipulated to meet desired needs. In addition, students will have hands-on experience to **implements and perform fundamental image processing algorithms** such as image manipulation, image filtering, image enhancement, image feature extraction, and image classification by utilising relevant tools and technologies.

Continuous Assessment



■ Assignment ■ Test ■ Practical Assessment



DESKTOP SOFTWARE

Jupyter Notebook



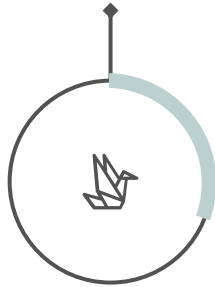
TS DR TAN CHI WEE

chiwee@tarc.edu.my

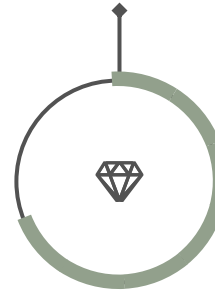


GROUND RULES

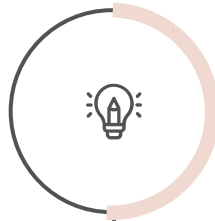
1
Respect each other



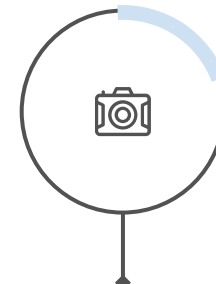
3
Use only positive language



2
Participate in the discussion: ask and
answer questions



4
Be punctual: assignment submission
and class attendance



CHAPTER 1

Introduction to Image Processing





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What is Digital Image Processing



02

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THANKS!

Coming up next...

Chapter 2
Image Filtering in Spatial Domain



01

INTRODUCTION

What is Digital Image Processing



What is Digital Image Processing?

- Image = Digital Image
- 2-dimensional function, $f(x,y)$
 - x and y = spatial (plane) coordinates
 - f = amplitude
- x, y, f = finite, **discrete** quantities
- The field of digital image processing = processing digital images by means of a digital computer



(a
)



(b
)

- 2-dimensional function, $f(x,y)$
 - x and y = spatial (plane) coordinates
 - f = amplitude
- What is amplitude, f in (a) & (b)?

Try make a
guess...



(a
)



(b
)

- x, y, f = pixels = used to denote the elements of a digital image.
- A pixel is also known as a picture element (pix = picture, el = element).
- A pixel is the smallest unit of a digital image or graphic that can be displayed and represented on a digital display device.

Pixels

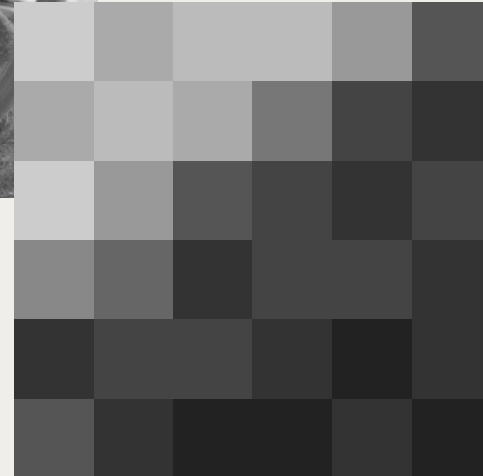


In **8-bit** representation
Pixel intensity values change between **0 (Black)**
and **255 (White)**

rows columns
 $f(645:650, 1323:1328) =$

83	82	82	82	82	82
82	82	82	81	81	81
82	82	81	81	80	80
82	82	81	80	80	79
80	79	78	77	77	77
80	79	78	78	77	77

$f(2724, 2336) = 88$





8-bit (256 tonal levels)



5-bit

Guess?



3-bit

Guess?

In **8-bit** representation
Pixel intensity values change between **0 (Black)**
and **255 (White)**



02

EXAMPLE

Fields that use Digital Image Processing



Robot Arms Selling Huawei Phones in New Unmanned Store

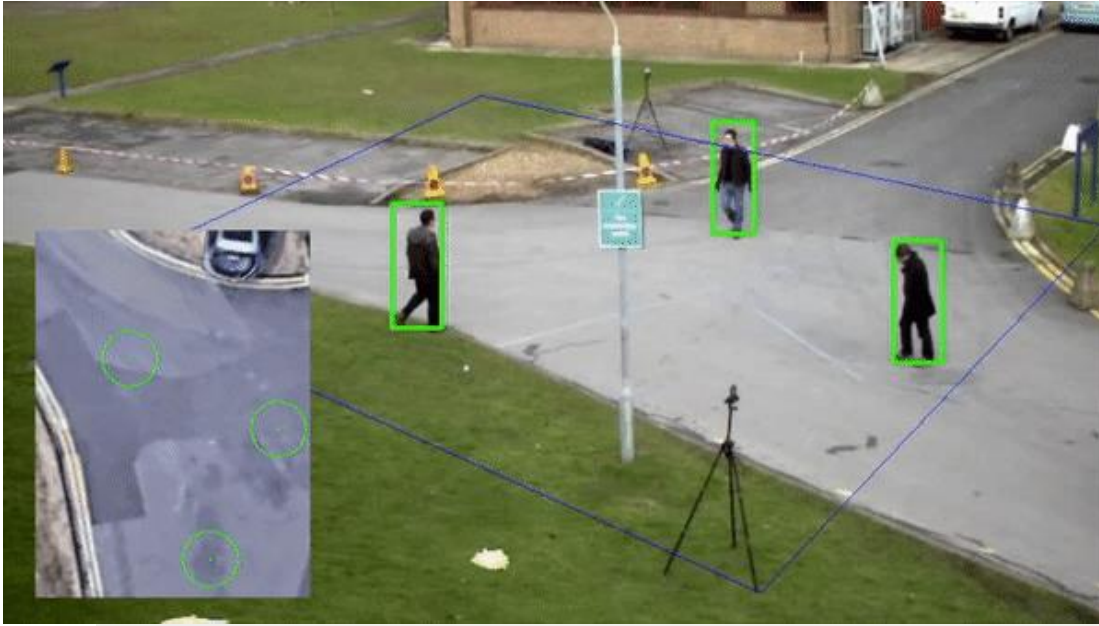
Computer Vision

- A kind of automated watchdog
- related to theory for design of artificial systems that acquire information from images
- Example of application:
 - industrial robot
 - autonomous vehicle
 - visual surveillance / people counting
 - interaction between computing machine and human



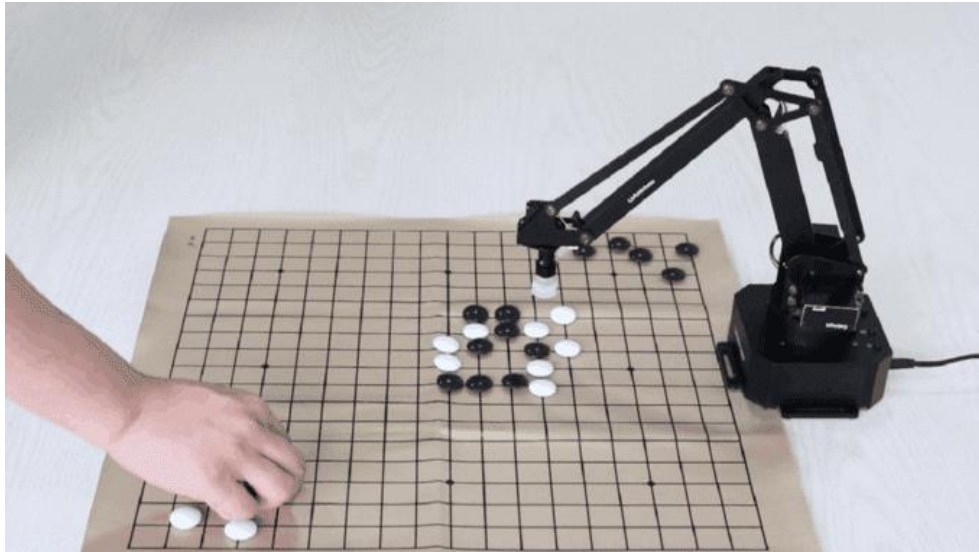
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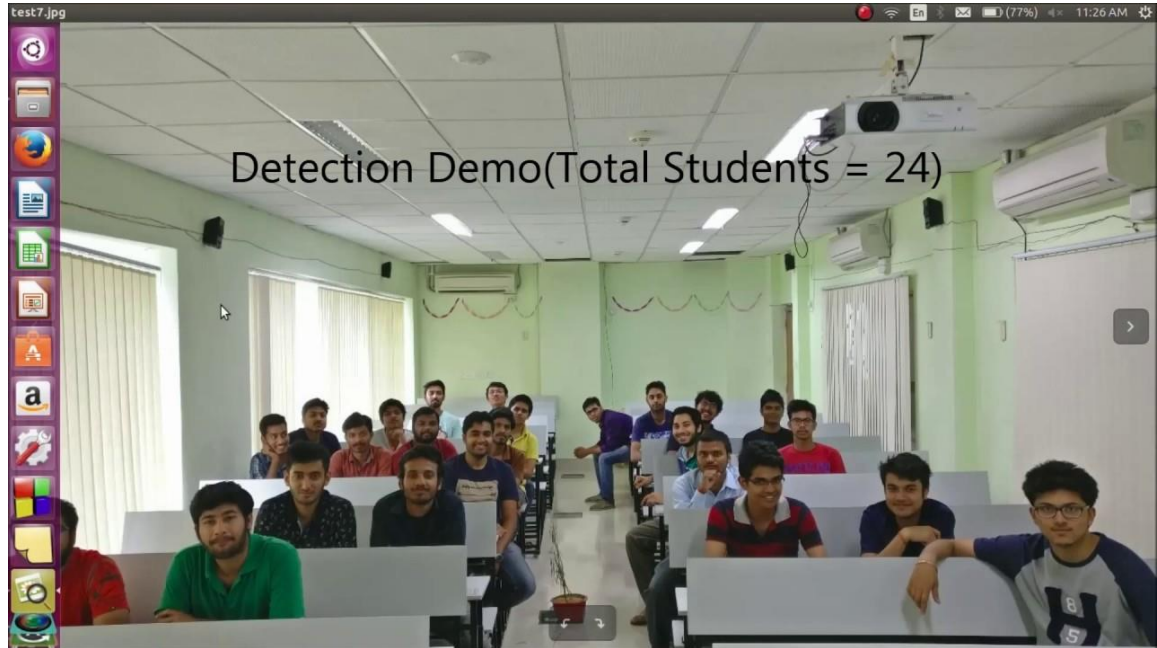
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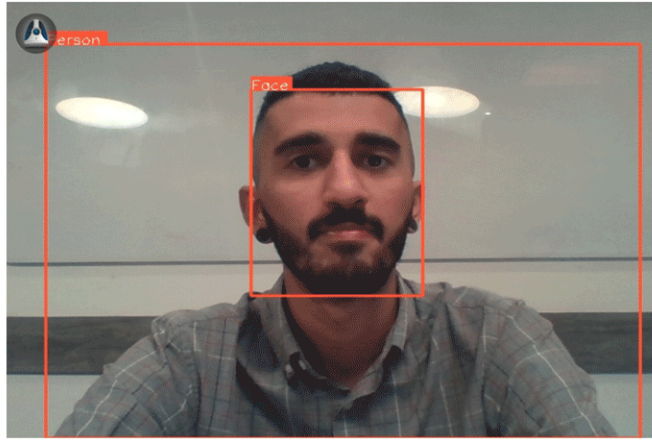
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Face Detection

- important facial features are detected and else are ignored.
- Face detection $\approx$ object class detection
- Various face detection algorithms are focused on the detection of frontal human faces.
- It is also an attempt to solve the more general and difficult problems of multi view face detection.



NEXT GENERATION PROCTORING -
Experience the real power of **AI**

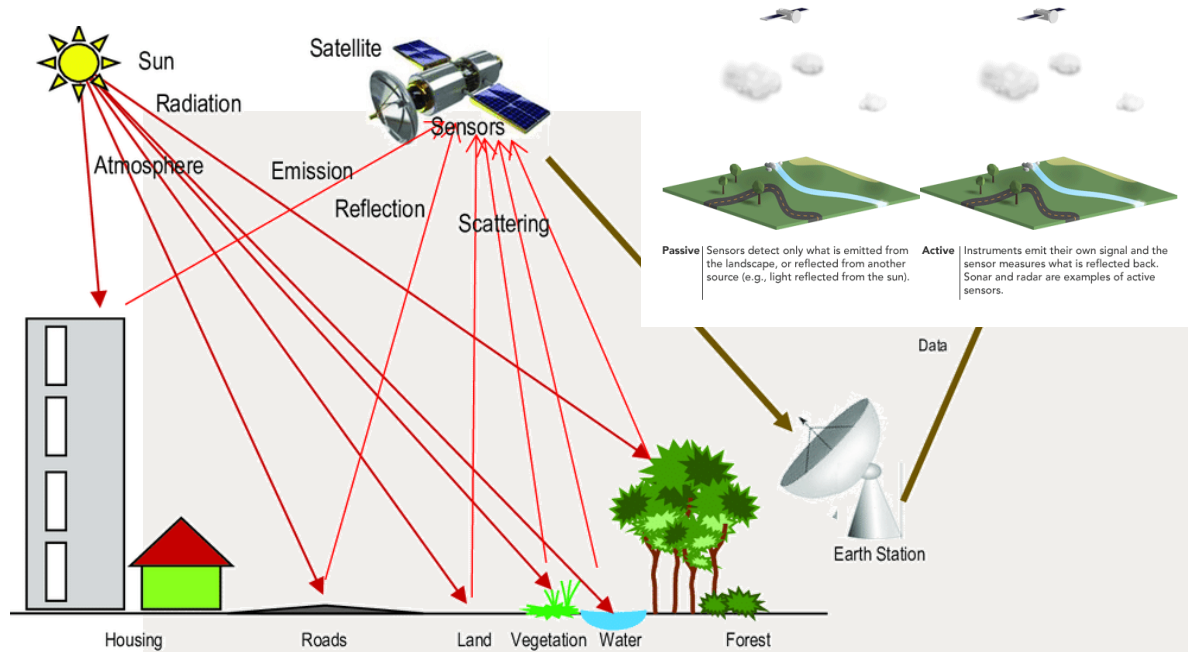
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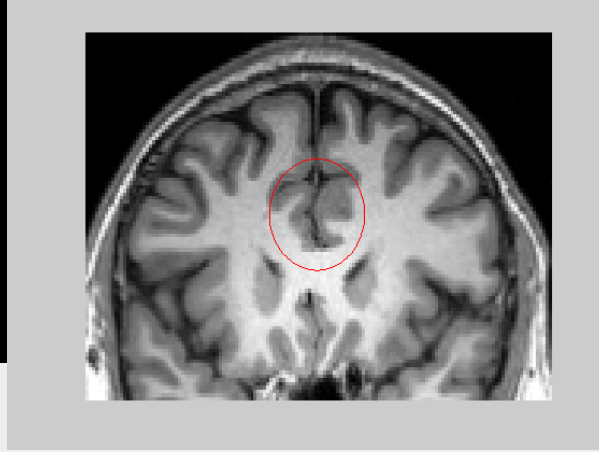
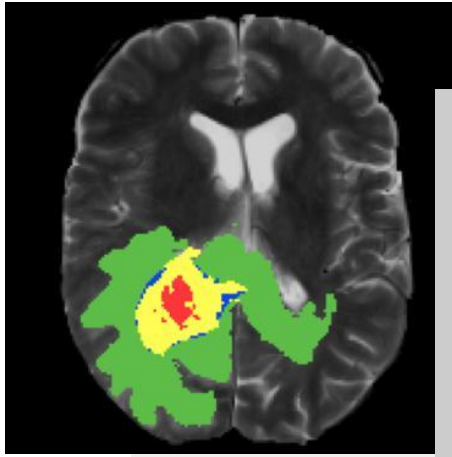
Digital Video Processing

- analyze images and videos as two- and three-dimensional signals in the spatial, spatio-temporal, and frequency domains.
- Video processing techniques are used in television sets, VCRs, DVDs, video codec, video players
- TikTok



Remote Sensing

- Remote sensing is basically an acquisition of small or large scale information signals from an object or phenomenon, by the using various real-time sensing devices that are wireless in nature, or not in physical or direct contact with the object (such as aircraft, spacecraft, satellite or ship)



Biomedical Image Enhancement & Analysis

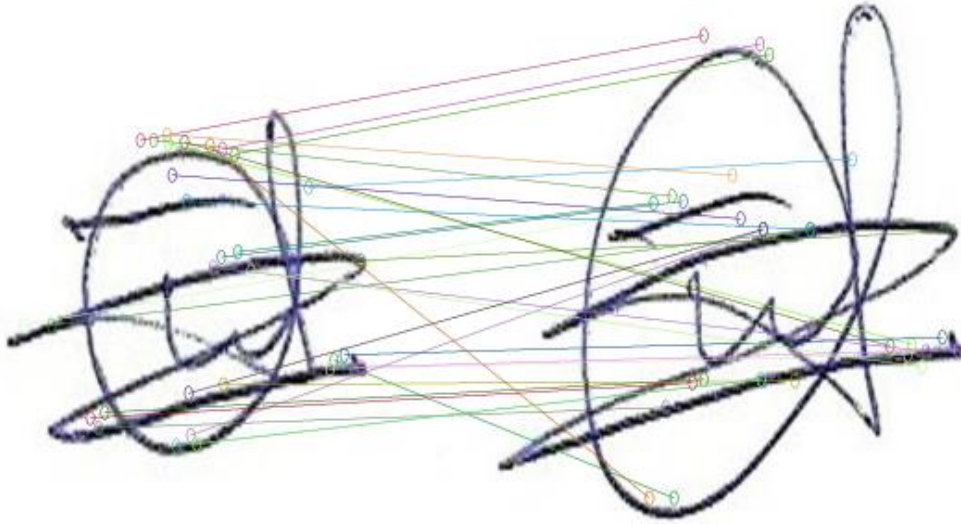
- enhance the biomedical images
- Computed Tomography (CT) or Magnetic Resonance Imaging (MRI), initially analog imaging modalities such as traditional applications like endoscopy or radiography are nowadays equipped with digital sensors.

DIGITAL WATERMARK



Watermarking & Steganography

- Steganography is used for secret communication
- Watermarking is used for content protection, copyright management, content authentication and tamper detection.



Signature Recognition

- to decide, whether a signature belongs to a given signer based on the image of signature and a few sample images of the original signatures of the signer
- different sizes and orientation in contrast to handwriting



(a)

(b)

(c)



Underwater Image Restoration & Enhancement

- When the light enters into water, it exponentially attenuates with the depth of water level; therefore the visibility distance is affected and so limited.
- Underwater images suffer from different problems such as blurring, non uniform lightening, noise, low contrast, etc.



Underwater Image Restoration & Enhancement

This computer vision algorithm removes the water from underwater images !

Form W-4
Department of the Treasury
Internal Revenue Service

Employee's Withholding Certificate

► Complete Form W-4 so that your employer can withhold the correct federal income tax.
► Give Form W-4 to your employer.
► Your withholding is subject to review by the IRS.

Step 1:
Enter Personal Information

(a) First name and middle initial
Adrian Rosebrock

(b) Address
PO Box 17598 #17900
Baltimore, MD 21297-1598

(c) ☐ Single or Married filing separately
☐ Married filing jointly (or Qualifying widow(er))
☐ Head of household (Check only if you're unmarried and pay more than half the costs of keeping up



Character Recognition

- Known as Optical Character Recognition (OCR)
- handwritten/printed text > machine editable text
- most effective and speedy method in terms of document input task.



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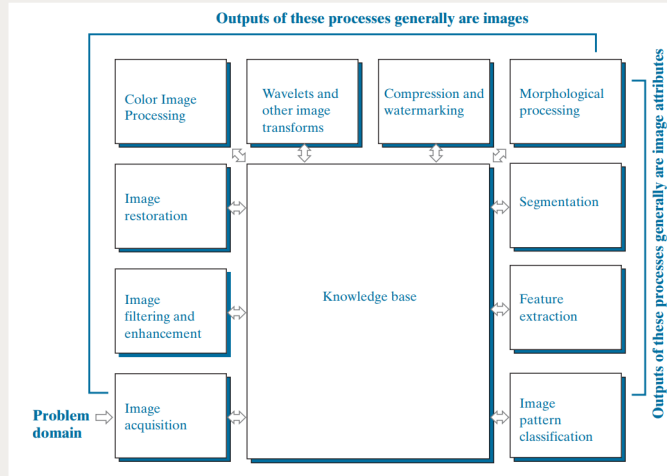


03

FUNDAMENTAL STEPS

Methods

Fundamental Steps in digital image processing



Outputs of these processes generally are images

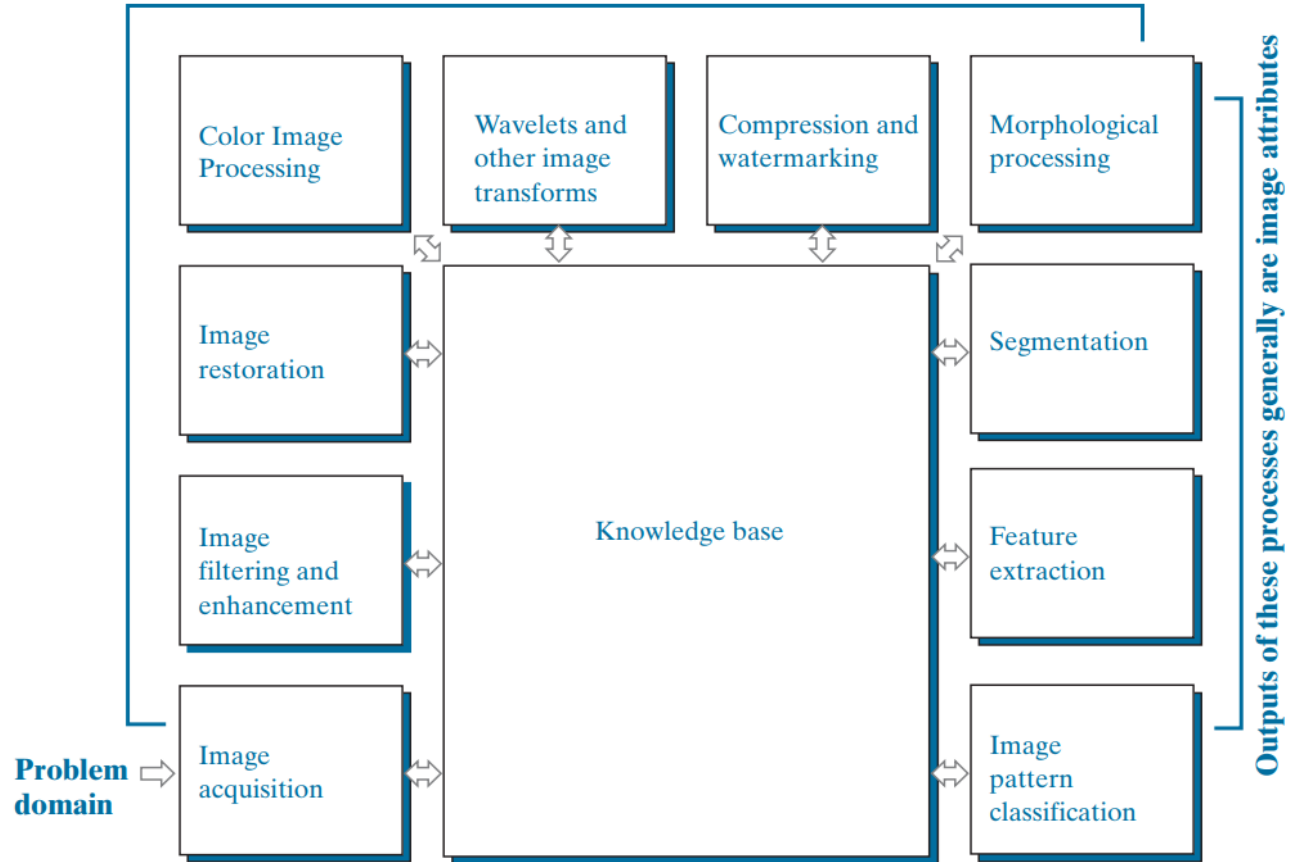




Image Filtering & Enhancement

Process of manipulating an image
so the result is more
suitable than the original for a
specific application

Image Filtering & Enhancement

No general “theory” of image enhancement



Image Restoration

Improving the appearance of
an image



Colour Image Processing

Fundamental concepts in colour model, basic for extracting features of interest in an image.





Image Compression

for reducing the storage required to
save an image, or the bandwidth
required to transmit it

Morphological Processing

the value of each pixel in the output image is based on a comparison of the corresponding pixel in the input image with its neighbours

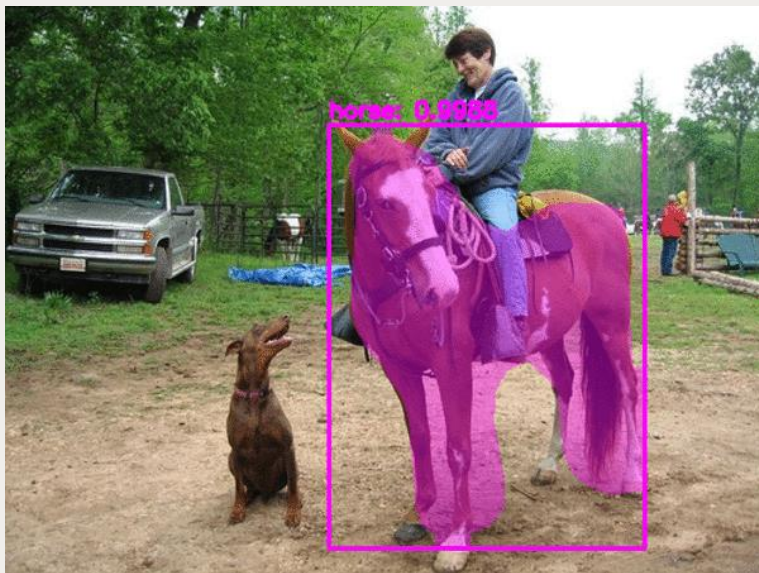
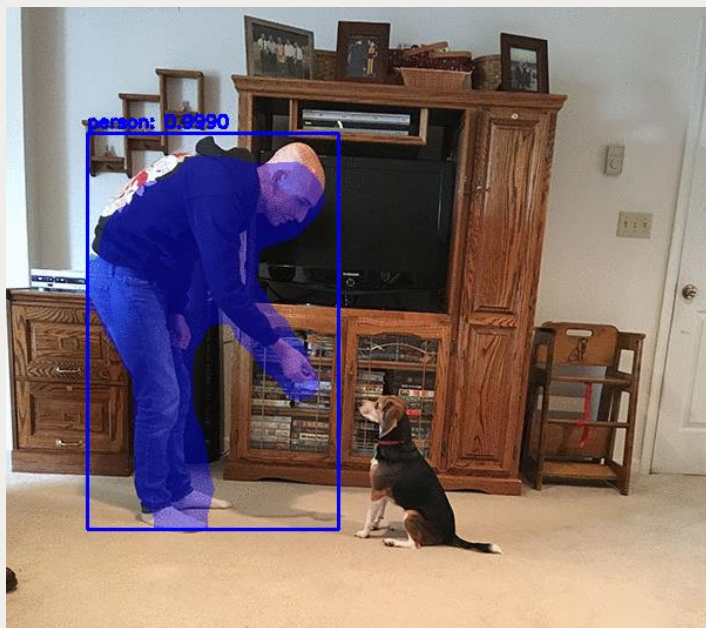




Segmentation

Image segmentation could involve separating foreground from background, or clustering regions of pixels based on similarities in colour or shape.





Mask R-CNN with OpenCV

Feature Extraction

Feature extraction consists of **feature detection** (to finding the features in an image, region, or boundary) and **feature description** (assigns quantitative attributes to the detected features).



Image Pattern Classification

Feature extraction consists of **feature detection** (to finding the features in an image, region, or boundary) and **feature description** (assigns quantitative attributes to the detected features).



Outputs of these processes generally are images

